Valentina Vecchio, PhD

Research Scientist: Data Scientist & Global group Leader

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Personal Website | GitHub | ORCID

PROFILE

I am an experienced and established scientific researcher with a passion for discovering and making sense of statistical patterns within huge data sets, and for finding ways to communicate their meaning and import, both to peer scientists and to laymen.

I have been a key participant in the ATLAS collaboration at CERN since 2015, an international and ongoing scientific experiment that this year was awarded the prestigious 2025 Breakthrough Prize. ATLAS' research contributed to the discovery of the Higgs Boson, a major advance recognised with a Nobel Prize.

I am a highly influential player in the collaboration, not only for my intellectual contributions to several high-profile analyses related to properties of the Higgs Boson, but also for my role as a facilitator resolving complex and sometimes conflicting objectives of different stakeholders and collaborating institutions, and for my leadership of several research teams.

PROFESSIONAL EXPERIENCE

Research Associate, University of Manchester
11/2019 – present | Manchester, UK

- I carry my research in experimental particle physics within the ATLAS collaboration at CERN. There I develop advanced data analyses for the study of the Higgs boson properties and the implementation of bleeding edge machine learning techniques for the identification of bottom-quark (flavour tagging).
- I have excelled in progressively more senior leadership roles, whilst also achieving ground-breaking scientific outcomes.
- I have mentored several PhD/MSc Students.
- I kick-started a UK-based effort to showcase the potential of Muon Colliders, where I lead the team developing a transformer-based flavour tagging algorithm to enable precise Higgs Boson measurements.

Associate Fellow, CERN
07/2018 – 06/2019 | Geneva, Switzerland

Summer Intern, Fermilab National Laboratory
07/2015 – 09/2015 | Batavia, Illinois (USA)

SKILLS

Data Analysis & Machine Learning	●●●●●
Leadership & Mentorship	●●●●●
Data processing & visualisation	●●●●●
Communication	●●●●●
Statistical Inference	●●●●●
Mathematical Thinking	●●●●●
Machine Learning & Deep Learning	●●●●○
Git | CI/CD | MkDocs	●●●●●
Python | C++ | Bash | LaTeX	●●●●○
Numpy | Pandas | Polars	●●●○○
Scikit-learn | PyTorch | XGBoost	●●●○○

LANGUAGES

Italian | English | Spanish | French

TALKS

Complete List

INTERESTS

- Fantasy Premier League
- Film Photography

LEADERSHIP AND RESEARCH

10/2024 – present — **Lead Convener of international Flavour Tagging programme**

- I currently lead a group of 200 researchers and PhD students across multiple institutions around the world. Under my leadership we have revolutionised the paradigm of jet flavour tagging, a key technique in the identification of bottom-quarks. We developed, deployed and calibrated the first ever transformer-based tagger, with a four-fold improvement in performance.

- I provide strategic guidance to the group to meet the collaboration's broader goal of driving innovative research and discovery. I liaise between senior stakeholders and team members to translate strategic priorities into achievable deliverables. This has sometimes required me to facilitate difficult conversations between stakeholders in order to resolve conflicting agendas. I design and assign tasks, working with the convenors of sub-groups, and communicate the aims of each task to management. I review the progress and outcomes of work in the sub-groups, and develop measurable benchmarks to assess analytical effectiveness.
- I lead the articulation and communication of compelling narratives to explain our scientific findings both to the wider research community, to funding stakeholders and to the public.

03/2021 – present

Lead Higgs Boson analyst
- I led the team that designed the first ever measurement of the Higgs boson coupling to one top-quark using the ATLAS detector. I am one of three editors of a paper shortly to be published documenting these results, and I am playing a key role in communicating and visualising the outcomes.
- I wrote a Python tool for the generation of a statistical inference model to harmonise the fitting strategy across the different channels of the analysis. I designed the analysis strategy for three of the five channels, employing BDTs for mitigation of the background contribution and both data-driven and MC-simulated studies of the background composition.

10/2022 – 09/2024

Convener of Flavour Tagging Calibration sub-group
- I led the first ever measurement of the performance of a transformer-based tagger, estimating from data a background subject to extremely high rejections (~1.8k).
- I identified and fixed a weak step in the deployment of recommendations to the collaboration. This required the development of a standalone plotting tool, written in Python, to cross-check consistency between calibration results in the deployment format, accelerating its validation process significantly.

03/2021 – 02/2023

Liaison lead for ATLAS in the UK
- For several years, I was the point of contact and liaison for all researchers in UK institutions working on top-quark physics as part of the ATLAS experiment.
- I organised annual workshops at which researchers could meet up, present their work, exchange ideas and establish new collaborations.

10/2020 – 09/2022

Convener of Flavour Tagging 'X->bb' sub-group
- I led the team that developed the first ever data-driven method for the calibration of a neural network tagger for the identification of highly energetic Higgs Boson objects.
- I conducted validation studies on the tagger, to understand its performance on Monte Carlo (MC) simulated events, and designed a novel data-driven method for the measurement of tagger performance. I developed a C++ code-base that allows the full chain of calibration to run with less steps and in less time. I pioneered the development of a simple Boosted Decision Tree algorithm to increase the purity of a very challenging calibration data-set.

EDUCATION

10/2016 – 10/2019
Rome/Geneva, Italy/Switzerland

PhD in Physics, Università degli Studi Roma Tre
- Designed a novel analysis for the statistical inference of the top-bottom quark coupling - achieving the highest single-measurement precision
- Improved and fastened the C++ software used to measure in data the performance of a Neural Network jet flavour tagger

10/2014 – 09/2016
Rome, Italy

Masters in Physics [110/110 Cum Laude], Università degli Studi Roma Tre
Wrote a BDT algorithm for the reconstruction of background events in the analysis that lead to the first ever observation of the associated production of the Higgs boson to two top-quarks

10/2011 – 09/2014
Rome, Italy

Bachelor in Physics [110/110 Cum Laude], Università degli Studi Roma Tre